

Academic Experience

- 2023–curIndependent Research Fellow, Kempner Institute, Harvard University, USA.
- 2018–2023Ph.D. Machine Learning, Advisor: Max Welling; University of Amsterdam, The Netherlands.
- 2015–2017M.S. Computer Science, Advisor: Garrison Cottrell; University of California San Diego, USA.
- 2011–2015B.S. Computer Science (Honors), Advisor: Yasser Abu-Mostafa; California Institue of Technology, USA.

Industry Experience

- 2022Research Intern, Apple Machine Learning Research. Barcelona, Spain.
- 2016–2018Deep Learning Data Scientist. Nervana → (acquired) → Intel AI Lab. San Diego, USA.

Teaching Experience

- 2025Course Instructor: "The Neuroscience of Artificial Intelligence". Harvard College, Undergraduate Seminar.
- 2020Teaching Assistant: "Machine Learning 1". University of Amsterdam, Bachelor's.
- 2019Teaching Assistant: "Machine Learning 2". University of Amsterdam, Master's.
- 2016Head Teaching Assistant: "Data Visualization". University of California San Diego, Master's.

Selected Advising Experience

- 2023–curEmma Finn. Undergraduate, Harvard College. (Daily supervisor)
Outcome: Rhodes Scholarship Recipient '25 & Two Kempner Undergraduate Research Fellowships.
- 2024–curMozes Jacobs. Ph.D. Candidate, Harvard. (Daily supervisor during primary advisor's sabatical; last author)
Outcome: CCN '25 Proceedings (Oral), & ICLR '25 (top 7% scores).
- 2025–curHansen Lillemark. Ph.D. Candidate, UCSD. (Daily supervisor for summer research visit; last author)
Outcome: ICLR '25 (under review), Accepted submission at 2 NeurIPS '25, Workshops .
- 2020–2021Qinghe Gao. Master's Candiate, University of Amsterdam. (Daily supervisor for master's thesis)
Outcome: Best Paper Award @ NeurIPS '25 SVRHM workshop.

Selected Publications

- NeurIPS '25 SpotlightT. A. Keller (2025). "Flow Equivariant Recurrent Neural Networks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. **Spotlight, Top 13% accepted**. arXiv: [2507.14793](https://arxiv.org/abs/2507.14793). url: <https://openreview.net/forum?id=N1KP0lcN6P>
- NeurIPS '25A. Karuvally, F. Nowak, T. A. Keller, C. A. Alonso, T. Sejnowski, and H. T. Siegelmann (2025). "Bridging Expressivity and Scalability with Adaptive Unitary SSMS". In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. url: <https://openreview.net/forum?id=s4zitEu2R8>
- ICLR '25 Under ReviewH. Lillemark, B. Huang, F. Zhan, Y. Du, and T. A. Keller (2025). "Flow Equivariant World Modeling for Partially Observed Dynamic Environments". In: *International Conference on Learning Representations (ICLR)*. Under review. url: <https://openreview.net/forum?id=W7WUJTGByR>
- ICLR '25 Under ReviewM. Jacobs, T. Fel, R. Hakim, A. Brondetta, D. E. Ba, and T. A. Keller (2025). "Block Recurrent Dynamics in Vision Transformers". In: *International Conference on Learning Representations (ICLR)*. Under review. url: <https://openreview.net/forum?id=gH3HhnfWLC>
- CCN '25 OralM. Jacobs, R. C. Budzinski, L. Muller, D. E. Ba, and T. A. Keller (2025). "Traveling Waves Integrate Spatial Information Through Time". In: *Conference on Cognitive Computational Neuroscience (CCN)*. **Oral presentation, Top 7%**. url: <https://openreview.net/forum?id=QEzqo546V5>

Springer '25 Book Y. Song, T. A. **Keller**, N. Sebe, and M. Welling (May 2025). *Structured Representation Learning*. en. Synthesis Lectures on Computer Vision. Cham, Switzerland: Springer International Publishing. url: <https://link.springer.com/book/10.1007/978-3-031-88111-4>

NeurIPS '25 Y. Song, T. A. **Keller**, S. Brodjian, T. Miyato, Y. Yue, P. Perona, and M. Welling (2025). "Kuramoto Orientation Diffusion Models". In: *Advances in Neural Information Processing Systems (NeurIPS)*. url: <https://arxiv.org/abs/2509.15328>

ICLR '25 Under Review J. Bertram, L. Dyballa, T. A. **Keller**, S. Kinger, and S. W. Zucker (2025). "How Neural is a Neural Foundation Model?" In: *Submitted to The Fourteenth International Conference on Learning Representations*. under review. url: <https://openreview.net/forum?id=jUy7vFgoZf>

AISTATS '25 Under Review E. L. Byrnes Finn, B. Wang, T. A. **Keller**, and D. E. Ba (2026). "Where the Score Lives: A Wavelet View of Diffusion". In: *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*. under review

PNAS '25 L. H. B. Liboni, R. C. Budzinski, A. N. Busch, S. Löwe, T. A. **Keller**, M. Welling, and L. E. Muller (2025). "Image segmentation with traveling waves in an exactly solvable recurrent neural network". In: *Proceedings of the National Academy of Sciences* 122.1, e2321319121. doi: [10.1073/pnas.2321319121](https://doi.org/10.1073/pnas.2321319121). eprint: <https://www.pnas.org/doi/pdf/10.1073/pnas.2321319121>. url: <https://www.pnas.org/doi/abs/10.1073/pnas.2321319121>

NeurIPS '25 S. Bharthulwar, T. A. **Keller**, M. Theodosis, and D. E. Ba (2025). "From Extrapolation to Generalization: How Conditioning Transforms Symmetry Learning in Diffusion Models". In: *NeurIPS 2025 Workshop on Symmetry and Geometry in Neural Representations*. url: <https://openreview.net/forum?id=UI82R3lwar>

arXiv '25 Y. Song, T. A. **Keller**, Y. Yue, P. Perona, and M. Welling (2025). *Langevin Flows for Modeling Neural Latent Dynamics*. arXiv: [2507.11531](https://arxiv.org/abs/2507.11531) [cs.LG]. url: <https://arxiv.org/abs/2507.11531>

ICLR '24 T. A. **Keller**, L. Muller, T. Sejnowski, and M. Welling (2024). "Traveling Waves Encode the Recent Past and Enhance Sequence Learning". In: *International Conference on Learning Representations (ICLR)*. url: <https://openreview.net/forum?id=p4S5Z6Sah4>

arXiv '24 T. A. **Keller**, L. Muller, T. J. Sejnowski, and M. Welling (2024). *A Spacetime Perspective on Dynamical Computation in Neural Information Processing Systems*. arXiv: [2409.13669](https://arxiv.org/abs/2409.13669) [q-bio.NC]. url: <https://arxiv.org/abs/2409.13669>

TPAMI '24 Y. Song, T. A. **Keller**, Y. Yue, P. Perona, and M. Welling (2024). "Unsupervised Representation Learning from Sparse Transformation Analysis". In: *arXiv preprint arXiv:2410.05564*. IEEE Transactions on Pattern Analysis and Machine Intelligence. doi: [10.48550/arXiv.2410.05564](https://doi.org/10.48550/arXiv.2410.05564). arXiv: [2410.05564](https://arxiv.org/abs/2410.05564) [cs.LG]. url: <https://arxiv.org/abs/2410.05564>

arXiv '24 E. Finn, T. A. **Keller**, E. Theodosis, and D. E. Ba (2024). *Learning Artistic Signatures: Symmetry Discovery and Style Transfer*. arXiv: [2412.04441](https://arxiv.org/abs/2412.04441) [cs.CV]. url: <https://arxiv.org/abs/2412.04441>

ICML '23 T. A. **Keller** and M. Welling (July 2023). "Neural Wave Machines: Learning Spatiotemporally Structured Representations with Locally Coupled Oscillatory Recurrent Neural Networks". In: *Proceedings of the 40th International Conference on Machine Learning (ICML)*. vol. 202. Proceedings of Machine Learning Research. PMLR, pp. 16168–16189. url: <https://proceedings.mlr.press/v202/keller23a.html>

NeurIPS '23 Y. Song, T. A. **Keller**, N. Sebe, and M. Welling (2023). "Flow Factorized Representation Learning". In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 36. Curran Associates, Inc., pp. 49761–49782. url: https://proceedings.neurips.cc/paper_files/paper/2023/file/9bfc2c20fa2f56a18397eafe1be8a50a-Paper-Conference.pdf

TMLR '23 T. A. **Keller**, X. Suau, and L. Zappella (2023). "Homomorphic Self-Supervised Learning". In: *Transactions on Machine Learning Research*. issn: 2835-8856. url: <https://openreview.net/forum?id=tEKqQgbwbf>

ICML '23 Y. Song, T. A. **Keller**, N. Sebe, and M. Welling (2023). "Latent traversals in generative models as potential flows". In: *Proceedings of the 40th International Conference on Machine Learning*. ICML'23. Honolulu, Hawaii, USA: JMLR.org

ICML '23 X. Suau, F. Danieli, T. A. **Keller**, A. Blaas, C. Huang, J. Ramapuram, D. Busbridge, and L. Zappella (2023). "DUET: 2D structured and approximately equivariant representations". In: *Proceedings of the 40th International Conference on Machine Learning*. ICML'23. Honolulu, Hawaii, USA: JMLR.org

- ICML '21 T. A. **Keller**, J. W. T. Peters, P. Jaini, E. Hoogeboom, P. Forré, and M. Welling (July 2021). "Self Normalizing Flows". In: *Proceedings of the 38th International Conference on Machine Learning (ICML)*. vol. 139. Proceedings of Machine Learning Research. PMLR, pp. 5378–5387. url: <https://proceedings.mlr.press/v139/keller21a.html>
- ICCV '21 Oral T. A. **Keller** and M. Welling (2021). "Predictive Coding with Topographic Variational Autoencoders". In: *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*. **Oral presentation**, pp. 1086–1091. doi: [10.1109/ICCVW54120.2021.00127](https://doi.org/10.1109/ICCVW54120.2021.00127)
- SVRHM '21 Best Paper Award T. A. **Keller***, Q. Gao, and M. Welling (2021). "Modeling Category-Selective Cortical Regions with Topographic Variational Autoencoders". In: *Shared Visual Representations in Humans and Machines (SVRHM) Workshop @ NeurIPS*. **Best Paper Award**. url: https://openreview.net/forum?id=yGRq_lW54bI
- NeurIPS '21 T. A. **Keller** and M. Welling (2021). "Topographic VAEs Learn Equivariant Capsules". In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 34. Curran Associates, Inc., pp. 28585–28597. url: <https://proceedings.neurips.cc/paper/2021/file/f03704cb51f02f80b09bffb15751691-Paper.pdf>

Invited Talks

- 2025 **NSF IAIFAI Colloquium** (MIT/Harvard, USA): "Flow Equivariance: Enforcing Time-Parameterized Symmetries in Sequence Models". 📺 Recording: youtube.com/watch?v=fO-wsJgGeL8
- 2025 **Fields Symposium on Spatiotemporal Neural Dynamics** (U of T, CA): "The Computational Inductive Biases of Spatiotemporal Artificial Neural Networks". 📺 Recording: youtube.com/watch?v=Rd-4lubEYc4
- 2025 **Neuro AI Emerging Leaders Webinar**: "The Computational Inductive Biases of Spatiotemporal Dynamics". 📺 Recording: youtube.com/watch?v=nmiGZo5uPH4
- 2024 **NeurReps International Speaker Series**: "A Spacetime Perspective on Neural Information Processing Systems". 📺 Recording: youtube.com/watch?v=FeVHfmRSicQ.
- 2024 **Active Inference Institute Seminar** (Virtual): "Natural Structure in Artificial Neural Networks". 📺 Recording: youtube.com/watch?v=1uIRljnLtc4.
- 2025 **AI, Brain, & Cognition Seminar, Center for NeuroImaging Research, Korea** (Virtual): "Computational Inductive Biases of Spatiotemporal Dynamics".
- 2025 **Guest Lecture: Applied Math 220, Geometric Deep Learning @ Harvard** (Harvard, USA): "Flow Equivariance".
- 2024 **Yale Wu Tsai Neuro AI Seminar**: "A Spacetime Perspective on Neural Information Processing Systems".
- 2024 **Woods Hole Computational Neuroscience Workshop** (Telluride, USA): "Wave Information Processing Systems".
- 2024 **CERN CMS Machine Learning Seminar** (Geneva, Switzerland): "Traveling Waves in Brains and Machines".
- 2024 **Computational Neuroscience Next Generation Symposium** (WashU, USA): "A Spacetime Perspective on Neural Information Processing Systems".
- 2023 **Frankfurt Institute for Advanced Studies, Giersch School and International Conference on Complex Systems** (Frankfurt, Germany): "Generalized Equivariance through Fluid Representations in Brains and Machines".
- 2023 **A.D. de Groot Cognitive AI Lecture** (Amsterdam, Netherlands): "Topographic Organization and Traveling Waves as Natural Representational Structure for Artificial Neural Networks".
- 2023 **Workshop on Structured Learning, Chalmers AI Research Center** (Gothenburg, Sweden): "Natural Neural Structure for Artificial Intelligence".
- 2022 **Seminar on Advances in Probabilistic Machine Learning** (Helsinki, Finland): "Topographic Variational Autoencoders Learn Equivariant Capsules".

Awards

- 2025 **Spotlight (Top 13% Accepted)**.
"Flow Equivariant Recurrent Neural Networks", [Keller, T. A.](#), NeurIPS '25.
- 2025 **Contributed Talk (Top 7% Accepted)**.
"Traveling Waves Integrate Information over Space", Jacobs, M., ..., [Keller, T. A.](#), CCN '25.
- 2025 **Top Reviewer Award**.
NeurIPS '25. Award Value: \$800.

- 2025 **Contributed Talk (Top 17%).**
"Flow Equivariant Cybernetics", Lillemark, ..., [Keller, T. A.](#), Champalimaud Foundation Research Symposium '25.
- 2025 **Travel Award.**
Chamalimaud Foundation. Amount: \$800.
- 2025 **FENS Brain Conference Travel Grant.**
FENS. Amount: \$800.
- 2024 **Contributed Talk (Top 10%).**
"Traveling Waves in State Space Models". [Keller, T. A.](#). From Neuroscience to AI Systems (NAISys) Conference.
- 2023 **Travel Award.**
COSYNE. Amount: \$800.
- 2022 **Travel Award.**
CIFAR Neuroscience of Consciousness Winter School. Award Value: \$1,500.
- 2021 **Contributed Talk (Top 10%).**
"Predictive Coding with Topographic VAEs". [Keller, T. A.](#) & Welling, M., Visual Inductive Priors Workshop, ICCV '21.
- 2021 **Best Paper Award.**
"Modeling Catagory-Selective Cortical Regions with TVAEs". [Keller, T. A.*](#), [Gao, Q.*](#), & Welling, M., Shared Visual Representations in Humans and Machines (SVRHM) Workshop; NeurIPS '21. Award Value: \$300.

Community Service

- 2025-cur **Workshop Co-organizer**, Teluride Neuromorphic Engineering Workshop on Spatiotemporal Dynamics
- 2025-cur **Speaker Series Organizer**, Foundation Role Models Speaker Series, CRISP Lab, Harvard
- 2025-cur **Reviewer**, ICLR Workshop Proposals
- 2025-cur **Reviewer**, TMLR Proceedings
- 2025-cur **Reviewer**, UAI Proceedings
- 2025-cur **Reviewer**, AAAI Proceedings
- 2025-cur **Reviewer**, CCN Proceedings
- 2024-cur **Reviewer**, NeurIPS Proceedings
- 2024-cur **Reviewer**, Neural Computation
- 2024-cur **Reviewer**, ICLR Proceedings
- 2024-cur **Reviewer**, AISTats Proceedings
- 2025-cur **Program Committee (Reviewer)**, Re-Align Workshop
- 2024-cur **Program Committee (Reviewer)**, NeuReps Workshop
- 2019-2022 **Board Member**, Inclusive AI, University of Amsterdam, ivi.fnwi.uva.nl/ellis/inclusive-ai/